Displacement and velocity

1

a $v(t) = 12t$

b i Yes ii No iii No iv Yes

c $48\ \text{m/s}$

2

a $v(t) = 9t^2 - 8t$

b $20\ \text{m/s}$

3

a $v(t) = 6t + 5$

b $29\ \text{m/s}$

4

a $v(t) = \dfrac{9}{\sqrt{t}}$

b $3\ \text{m/s}$

5

a $67.5\ \text{m/s}$

b $v(t) = \dfrac{3t^2}{2} + 6t$

c $67.5\ \text{m/s}$

d $t = 2\ \text{s}$

6

a $0\ \text{m/s}$

b $(-9.8x + 11.27)\ \text{m/s}$

c $11.27\ \text{m/s}$

d $5.635\ \text{m/s}$

e $0\ \text{m/s}$

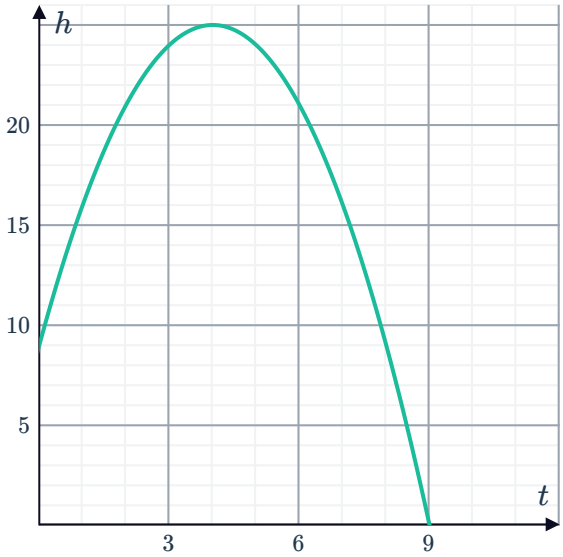
7

a $9\ \text{m}$

b $h = 25\ \text{m}$

c $t = 9\ \text{s}$

d



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8

a $h = 16\ \text{m}$ **b** $t = 6\ \text{s}$ **c** $\dfrac{dh}{dt} = -32\ \text{m/s}$

9

a $-4\ \text{m}$ **b** $v\left(t\right) = 6t^2 - 14t + 3$

c $3\ \text{m/s}$ **d** $1\ \text{m/s}$

10

a $9\ \text{m}$ **b** $v\left(t\right) = 32t + 24$ **c** $24\ \text{m/s}$ **d** $24\ \text{m/s}$

11

a $0\ \text{m}$ **b** $t = 4\ \text{s}$ **c** $40\ \text{m}$

12

a $0\ \text{m}$ **b** $v\left(t\right) = -4t - 20$ **c** $-20\ \text{m/s}$ **d** $20\ \text{m/s}$

13

a $\dfrac{ds}{dt} = 12 - 9.4t$ **b** $\dfrac{d^2s}{dt^2} = -9.4\ \text{m/s}^2$

14

a $\dfrac{dh}{dt} = -13\ \text{m/s}$ **b** $\dfrac{d^2h}{dt^2} = -10\ \text{m/s}^2$

15 $a = 1, b = 8$

Acceleration

16

a $v\left(t\right) = 4t - 4$ **b** $t = 1\ \text{s}$ **c** $a\left(t\right) = 4$

17

a $v\left(t\right) = 2t - 7$ **b** $-7\ \text{m/s}$ **c** $a\left(t\right) = 2$ **d** $2\ \text{m/s}^2$

e $t = 5\ \text{s}$ **f** $t = 5, 2\ \text{s}$

18

a $\dfrac{d^2s}{dt^2} = t + \dfrac{1}{2}$ **b** $\dfrac{1}{2}\ \text{m/s}^2$ **c** $\dfrac{33}{2}\ \text{m/s}^2$

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19

a $\dfrac{d^2s}{dt^2} = 12t^2 + 2$ **b** $2\ \text{m/s}^2$ **c** $110\ \text{m/s}^2$

20

a $a = 38 - 12t$ **b** $-10\ \text{m/s}^2$

c The particle is slowing down.

21

a $t = 4\ \text{s}$

b $0\ \text{m/s}$

c $t = 2\ \text{s}$

d $12\ \text{m/s}$

e The velocity is constant. The particle is neither speeding up or slowing down.

22

a $t = \dfrac{3}{2}\ \text{m/s}$

b $a\,(2) = 20\ \text{m/s}^2$

c The particle starts moving to the right, but eventually comes to rest. After instantaneously coming to rest, it then continues to move to the right.

23 $v = -\dfrac{25}{2}\ \text{m/s}$

24 $x = -12\ \text{m}$

25

a $t = 8\ \text{s}, 3\ \text{s}$ **b** $a\,(t) = 2t - 11$ **c** $-7\ \text{m/s}^2$

26

a $t = \dfrac{8}{5}\ \text{s}, 3\ \text{s}$ **b** $a\,(t) = 10t - 23$ **c** $7\ \text{m/s}^2$

27

a $t = 2$ **b** $x''\,(t) = -6t - 2$

c $x''\,(2) = -14\ \text{m/s}^2$ **d** $4\ \text{m/s}$

28

a $d' = 1.72t - 0.0252t^2$ **b** $t = 68\ \text{s}$

c $1335\ \text{m}$ **d** $d' = 29\ \text{m/s}$

e $d'' = 1.72\ \text{m/s}^2$

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29

a $\frac{9}{10}\,\text{s}$ **b** $s'(t) = u + at$ **c** $t = -\frac{u}{a}$ **d** $a = -\frac{50}{9}\,\text{m/s}^2$

Graphs of displacement, velocity and acceleration

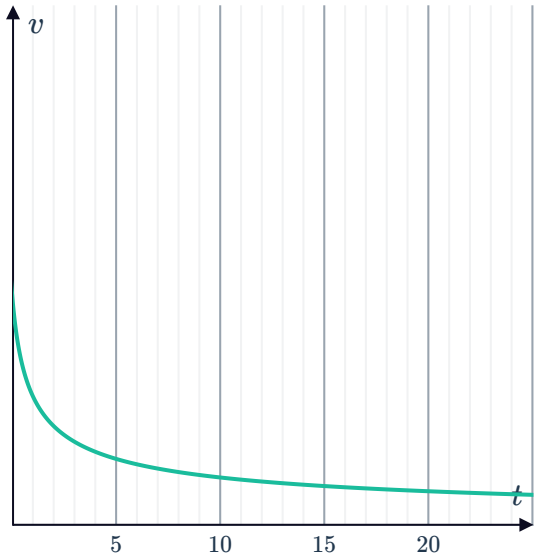
30

a $0\,\text{m}$ **b** $30\,\text{m}$

31

a $-6\,\text{m}$ **b** $6\,\text{m}$

32



33

a $3, 5, 10$ **b** $0 < t < 4, t > 8$ **c** $4, 8$

34

a $6, 11$ **b** $6 < t < 11$

35

a **i** $-2\,\text{m}$ **ii** $8\,\text{m}$ **b** $-3\,\text{m/s}$ **c** $2\,\text{m/s}$

36

a 2s **b** $0 < t < 2$ **c** $4\,\text{m}$ **d** $x = -t(t-4)$

e $4\,\text{s}$ **f** $\frac{dx}{dt} = -2t + 4$ **g** $-2\,\text{m/s}$

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37

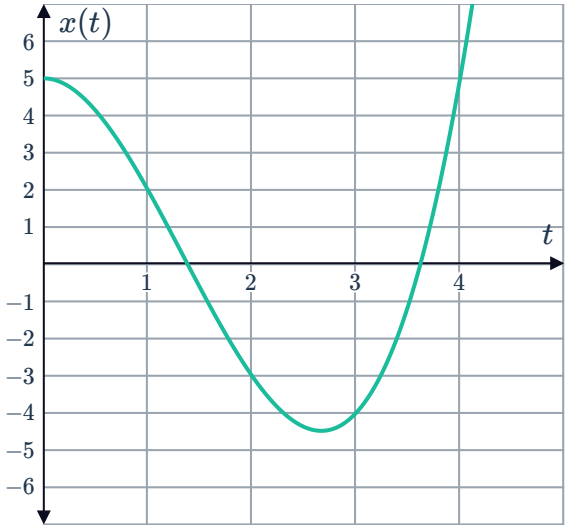
a 9 m **b** $6\text{ s}, 15\text{ s}$ **c** To the left **d** 11 s

e $-\frac{8}{5}\text{ m/s}$

38

a $v(t) = 3t^2 - 8t$ **b** $t = 0\text{ s}, \frac{8}{3}\text{ s}$

c



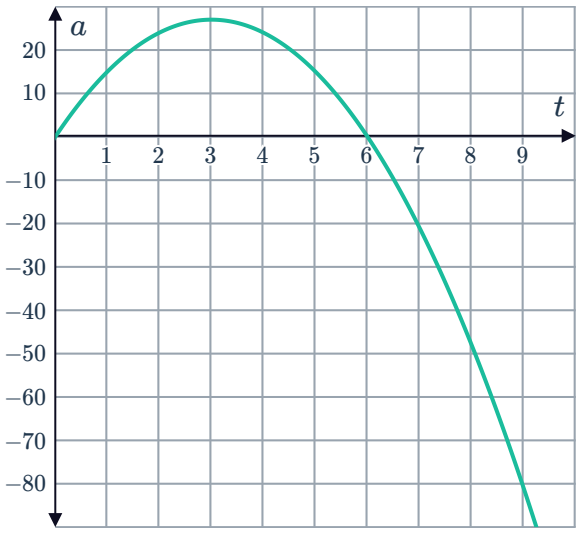
d $\frac{256}{27}\text{ units}$

39

a $a = 18t - 3t^2$ **b** $t = 6\text{ s}$

c 108

d



e 81

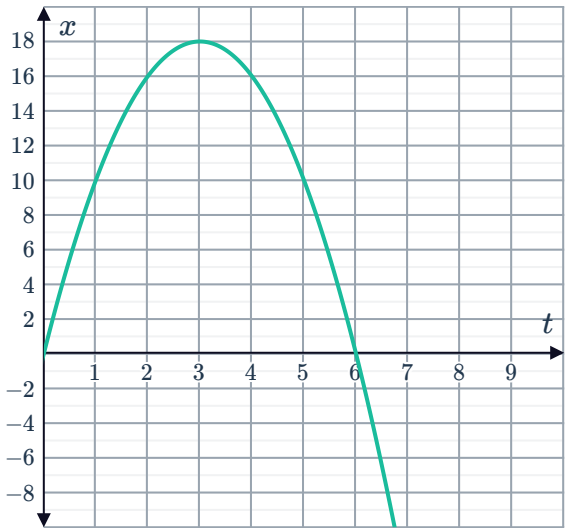
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40

a $v = 12 - 4t$

b $t = 3$ s

c



d -54 m

e 90 m

41

a $x'(t) = 3t^2 - 18t + 24$

b $x''(t) = 6t - 18$

c

i

$+$	$t < 2$
0	$t = 2$
$-$	$2 < t < 4$
0	$t = 4$
$+$	$t > 4$

ii

$-$	$t < 3$
0	$t = 3$
$+$	$t > 3$

d 2 s, 4 s

e $t < 2, 3 < t < 4$

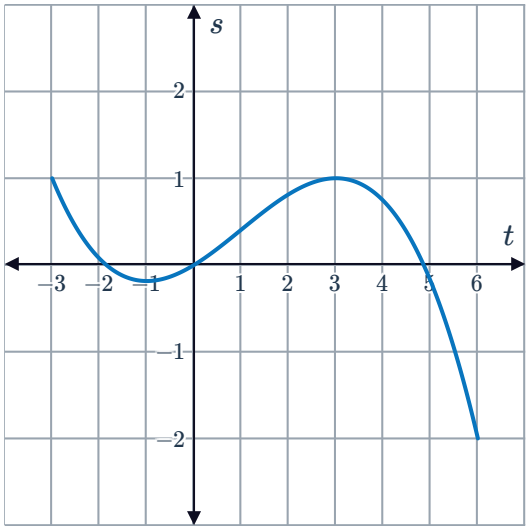
f $t < 3$

g 28 m

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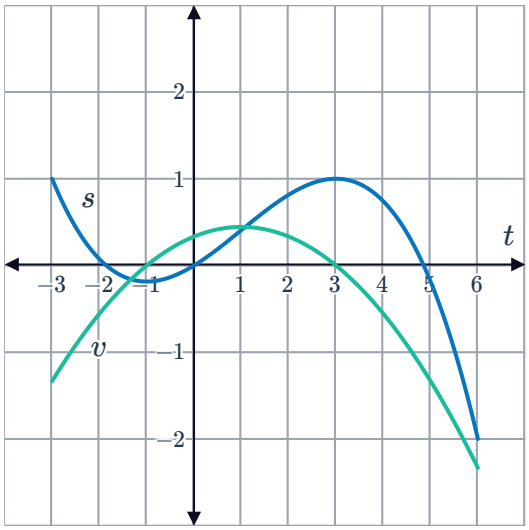
42

a



b $\frac{ds}{dt} = \frac{1}{3} + \frac{2}{9}t - \frac{1}{9}t^2$

c



d 4 s

e $-\frac{5}{27}\text{ m}$

f 1 m

g $\frac{8}{27}\text{ m/s}$

Displacement and acceleration from velocity

43

a

i $0 \leq t \leq 10$

ii No values of t.

b

i No values of t.

ii $0 \leq t \leq 10$

c

i $0 \leq t \leq 4, 8 \leq t \leq 10$

ii $4 \leq t \leq 8$

d

i $7 \leq t \leq 10$

ii $0 \leq t \leq 5$

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44

a

i 11 units

ii 19 units

b

i 6 units

ii 18 units

c

i -3 units (or 3 units to the left)

ii 15 units

d

i 1.75 units

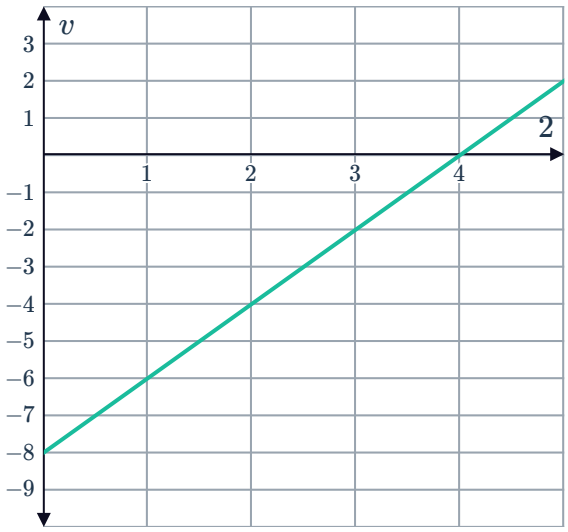
ii 19.75 units

45 Example answers:

- Because both are equal to the area of the triangle above the horizontal axis from $t = 0$ to $t = 6$.
- Because the particle only moved in one direction during the first six seconds.

46

a



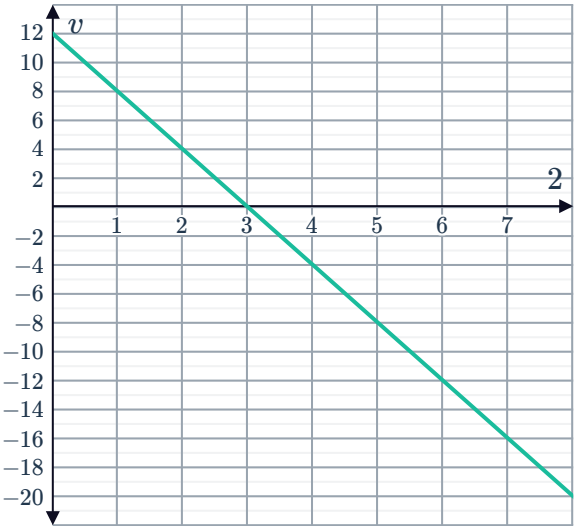
b -15 units or 15 units to the left.

c 17 units.

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47

a

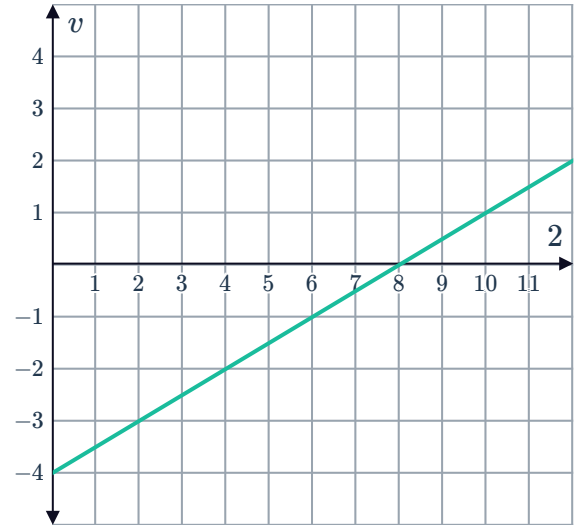


b -10 units or 10 units to the left.

c 26 units.

48

a



b -3 units or 3 units to the left.

c 5 units.

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